A High-Performance Rule-Based Multi-Agent Framework for Spatio-Temporal Vector Dynamics: Application to *Anopheles stephensi* in Ethiopia

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Abstract

Background: The invasive malaria vector *Anopheles stephensi* threatens urban Africa, but data scarcity limits conventional modelling. **Methods:** We developed a modular agent-based framework coupling a temperature-dependent Metzler matrix life-cycle model with spatial environmental layers (population, buildings, climate), behavioural rules, and a lazy spatial grid. The model was parameterised for *An. stephensi* and run for six months over two Ethiopian regions (Dire Dawa and Oromia). **Results:** Populations reached near-equilibrium ($\lambda_{\max} = 1.003$-1.074) with stage durations matching laboratory data. Urban cores showed highest suitability (Continuous Boyce Index up to 0.633). Adult abundance correlated weakly with temperature ($r = -0.12$, $p = 0.12$) but significantly with precipitation in Dire Dawa ($r = -0.72$, $p < 0.001$). **Conclusion:** The framework reproduces known *An. stephensi* ecology and identifies urban hotspots. Its YAML-driven (YAML is a human-readable, lightweight data serialization language used primarily for configuration files, data exchange, and DevOps tools like Kubernetes and Ansible) parameterisation makes it adaptable to other species and regions, supporting evidence-based vector control in data-limited settings.

Keywords: multi-agent systems, vector-borne disease, *Anopheles stephensi*, Metzler matrix, high-performance computing, spatial ecology.

1 Introduction

Vector-borne diseases (VBDs) represent one of the most profound challenges to global health security, accounting for approximately 700,000 deaths annually and imposing significant economic burdens on developing nations (Lowe and Codeço, 2025). Malaria alone threatens nearly half the world's population, with Sub-Saharan Africa bearing 94% of the global case burden as (Organization, 2024). While the past two decades witnessed substantial reductions in malaria mortality through coordinated interventions, this hard-won progress faces unprecedented threats from ecological and demographic shifts. The rapid urbanization of Sub-Saharan Africa, currently the world's fastest urbanizing region (Sulemana et al., 2019), has created novel transmission ecologies that challenges historical assumptions about vector ecology and control strategies (Joseph et al., 2024).

The primary driver of this epidemiological transition is the globalization of previously localized vectors, particularly the invasive malaria mosquito *Anopheles stephensi*. Unlike endemic African malaria vectors that predominantly breed in natural water bodies, *An. stephensi* exhibits remarkable ecological plasticity, thriving in artificial containers and man-made structures characteristic of urban environments (Hunter et al., 2025). This species shares critical adaptive traits with *Aedes aegypti*, the primary dengue vector, including container breeding, anthropophily, and exophilic resting behaviors (Hamlet et al., 2022). The consequences of this adaptation were first recorded in Djibouti, where *An. stephensi* establishment triggered a 30-fold increase in malaria cases within seven years, transforming a nation on the verge of malaria elimination into a high-transmission zone (Hamlet et al., 2022). This vector's capacity to exploit previously low-risk urban settings necessitates fundamentally new approaches to surveillance, modeling, and intervention design (Mwema et al., 2024).

A critical bottleneck in responding to this emerging threat is the pervasive "Data Gap" characterizing many resource-constrained settings where *An. stephensi* is establishing. Effective vector control relies on robust decision-support systems built upon high-quality, high-resolution entomological and epidemiological data, yet national surveillance programs often operate with limited financial and technological resources (Lowe and Codeço,

2025). Current modeling approaches struggle to bridge the gap between theoretical elegance and practical utility: empirical models propagate measurement errors through unreliable data streams, while process-based models frequently lack the spatial granularity and computational efficiency needed for operational deployment (Lowe and Codeço, 2025). This disconnect is particularly acute for invasive vectors, where absence of historical baseline data undermines traditional statistical approaches (Pascoe et al., 2019).

To address these challenges, we propose a novel computational framework that integrates high-performance computing methodologies with ecological process modeling. Our system implements a layered and modular, rule-based Multi-Agent Simulation (MAS) architecture that harmonizes heterogeneous spatio-temporal data streams into a unified analytical environment. This framework is part of the family of ABS (Agent Based Simulation) which are simulation models where the researcher explicitly describe the decision processes of simulated actors at micro-level (Siebers and Aickelin, 2008). The proposed framework bridges Computer Science and Environmental Science by conceptualizing environmental data as dynamic information layers, including static rasters (for instance: elevation, building density, population density), time-series climate variables, and vector data (shapefiles), that interact with autonomous agent populations. We distinguish between two agent classes: "Inert Agents" representing static environmental features (urban water containers serving as breeding sites) and "Living Agents" modeling biological vectors (*An. stephensi* mosquitoes for example) with behavioral rules derived from empirical studies.

This study demonstrates how such synthetic modeling approaches can provide critical insights where field data are scarce or unreliable. Using Dire Dawa, and Oromia woredas from Ethiopia, which are regions experiencing *An. stephensi* invasion, (Zhou et al., 2024; Hawaria et al., 2023) as a proof-of-concept, we simulate 6 months of mosquito population dynamics across the whole regions. Our results illustrate how agent-based modeling can identify high-risk micro-habitats, simulate seasonal population trajectories, and potentially assist intervention strategies in data-limited settings. By making the simulation framework open-source and computationally efficient, we aim to empower public health agencies in resource-constrained regions with decision-support tools capable of addressing the unique challenges posed by urban-adapted invasive vectors.

2 Materials and Methods

2.1 Framework Architecture

2.1.1 Overview

The simulation framework follows a modular, agent-based design that integrates spatially explicit environmental data, temperature-dependent life-cycle models, and behavioural rules. It is implemented in Java and built around four core subsystems: (i) a time manager that advances simulation time in discrete ticks (15-minutes per tick) and interpolates climate data; (ii) a spatial registry that indexes agents on a lazy grid for efficient neighbourhood queries; (iii) an agent layer system that manages agents and evaluates rules in parallel; and (iv) a lifecycle model that computes temperature-driven development and survival rates. The overall architecture is illustrated in Figure 1.

2.2 Core Components

2.2.1 Agent layers

Two agent types are implemented: `LivingAgent` (representing a mosquito) and `InertAgent` (representing a breeding site). Living agents track life stage (egg, larva, pupa, adult), age, energy, gravid status, and resting behaviour. Their movement is modelled as a random walk with a step size derived from field observations of flight range (Sayeed Quraishi et al., 1966). Inert agents maintain water volume, egg count, larval count, and carrying capacity; they update their capacity dynamically based on water level, mimicking density-dependent mortality observed in larval habitats (Thomas et al., 2016). Agent layers are processed in parallel using a partition-based container (`AgentContainer`) and a rule engine that evaluates behavioural rules (e.g., feeding, oviposition, resting) in priority order (Ogunlade et al., 2023).

2.2.2 Raster layers and climate data

Static raster layers (elevation, building density, population density) are loaded from GeoTIFF files using GeoTools and stored either in memory or memory-mapped for large grids. Climate time series (2-m temperature and total precipitation) are ingested from NetCDF files (ERA5-Land reanalysis) into interpolated raster layers that perform linear temporal interpolation between hourly steps (Chabot-Couture et al., 2014; Merkord et al., 2017).

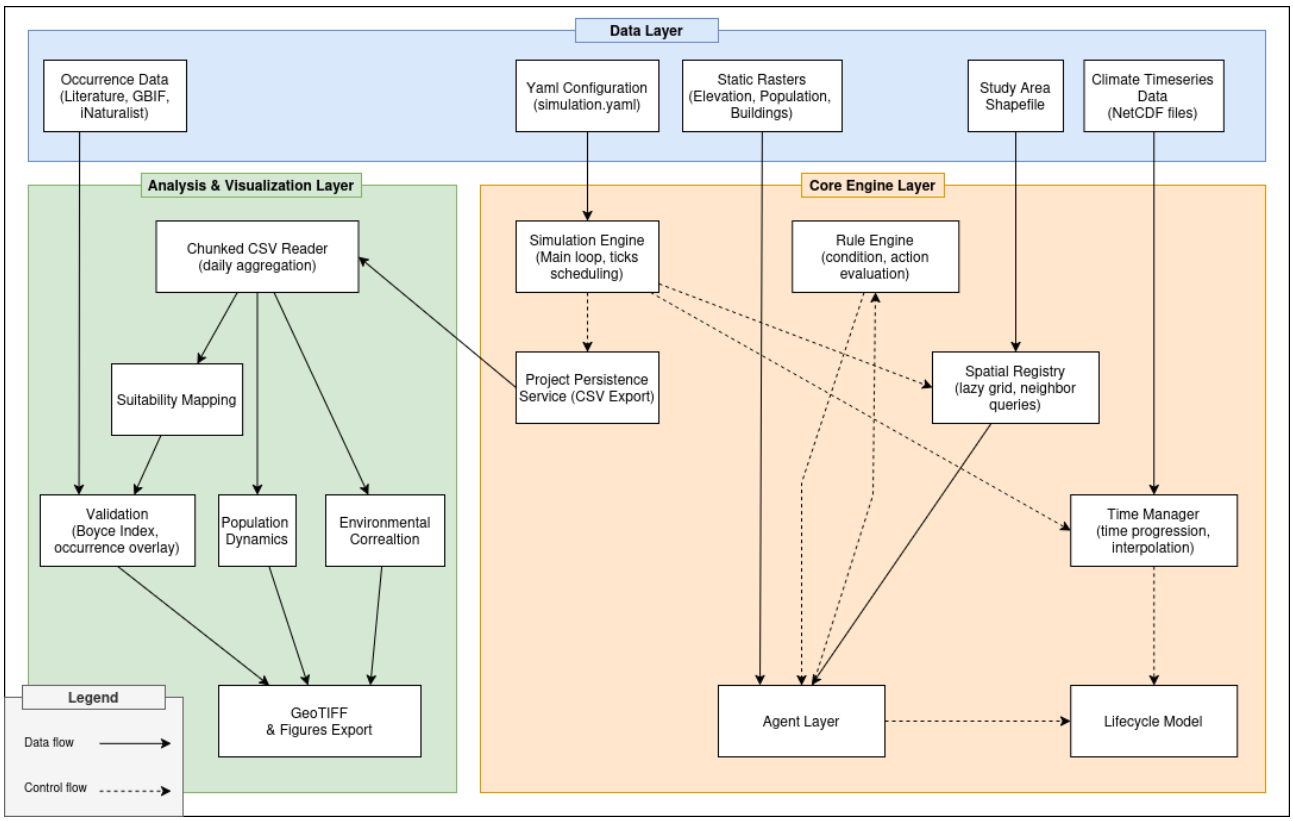


Figure 1: High-level architecture of the simulation framework. The diagram is organised into three layers: *Data Layer* (input data sources), *Core Engine Layer* (simulation components), and *Analysis & Visualization Layer* (post-processing and validation). Solid arrows denote *data flow* (e.g., configuration files, rasters, climate NetCDF, and simulation outputs). Dashed arrows denote *control flow* (e.g., tick scheduling, spatial queries, rule evaluation, and lifecycle updates). The analysis layer reads CSV snapshots generated by the persistence service and produces daily summaries, suitability maps, validation metrics, and final figures.

2.2.3 Lifecycle model

The lifecycle model represents mosquito population dynamics using a discrete-time, stage-structured Metzler matrix (Ogunlade et al., 2023; Agboka et al., 2025b). Four developmental stages are tracked: eggs (E), larvae (L), pupae (P), and adults (A). All rates are temperature-dependent and defined on a per-day basis, then converted to tick-level probabilities via the Poisson process:

$$p = 1 - \exp(-r\Delta t),$$

where Δt is the tick duration in days ($\Delta t = 0.25$ h $= 0.01042$ days). The daily projection matrix is:

$$\mathbf{M}(T) = \begin{pmatrix} (1-d_E)S_E & 0 & 0 & F \cdot K \\ d_E S_E & (1-d_L)S_L & 0 & 0 \\ 0 & d_L S_L & (1-d_P)S_P & 0 \\ 0 & 0 & d_P S_P & 1-\mu_A \end{pmatrix},$$

with stage abundances evolving as $[E, L, P, A]_{t+1}^\top = \mathbf{M}(T)[E, L, P, A]_t^\top$. The temperature-dependent functions are:

- **Development rates** (day^{-1}): $d_X(T) = \max(0, a_X T^2 + b_X T + c_X)$ for $X \in \{E, L, P\}$.
- **Stage survival** (Gaussian): $S_X(T) = \text{amp}_X \cdot \exp\left(-\frac{1}{2}\left(\frac{T-\mu_X}{\sigma_X}\right)^2\right)$.
- **Adult mortality** (day^{-1}): $\mu_A(T) = \max(0, a_A T^2 + b_A T + c_A)$.
- **Fecundity** (eggs/female/day): $F(T) = \max\left(0, a_F e^{b_F T} - e^{b_F T_{\max} - \left(\frac{T_{\max}-T}{c_F}\right)^2}\right)$.
- **Host carrying capacity**: $K = 1 + \ln(1 + \text{population density})$, which scales fecundity based on human population density (diminishing returns at high density).

- **Density-dependent larval mortality**: each aquatic habitat has a carrying capacity cap (linearly scaled with water volume); if larval count exceeds cap, excess larvae are removed probabilistically.

All temperature-response coefficients are stored in a YAML configuration file under the `species` key, making the model generic and easily adaptable to any mosquito species by replacing the coefficients in Table 1. This parametrisation follows the principles of generic insect population modelling (Agboka et al., 2025a,b, 2026).

Table 1: Generic temperature-response parameters for a mosquito lifecycle model.

Parameter set	Meaning	Unit	YAML key
d_E coeff.	Egg development rate	$1/°C^2, 1/°C, 1$	`egg_dev_a,b,c`
d_L coeff.	Larva development rate	$1/°C^2, 1/°C, 1$	`larva_dev_a,b,c`
d_P coeff.	Pupa development rate	$1/°C^2, 1/°C, 1$	`pupa_dev_a,b,c`
S_E	Egg survival (amp, mean, sigma)	dim., °C,° C	`egg_survival_...`
S_L	Larva survival (amp, mean, sigma)	dim., °C,° C	`larva_survival_...`
S_P	Pupa survival (amp, mean, sigma)	dim., °C,° C	`pupa_survival_...`
μ_A coeff.	Adult mortality	$1/°C^2, 1/°C, 1$	`adult_mort_a,b,c`
F coeff.	Fecundity	$a_F, b_F, T_{\max}, c_F$	`fecundity_a,b,Tmax,c`
K	Host scaling factor	dimensionless	(derived)

Note: "coeff." denotes the three coefficients (a, b, c) of the quadratic function. "1" indicates dimensionless (unit 1). "dim." stands for dimensionless. For survival parameters, "amp" is dimensionless, "mean" and "sigma" are in °C.

2.2.4 Rule engine and spatial registry

Behavioural rules (e.g., "`feed`", "`lay_eggs`", "`rest_in_building`") are defined in a YAML configuration file and evaluated using a JavaScript engine (GraalVM (Vergilio et al., 2023)). The spatial registry implements a lazy grid with a configurable cell size (default $0.001° \approx 110$ m). Neighbour queries are performed by checking cells within a Chebyshev distance $\lceil \text{radius}/\text{cellSize} \rceil$; results are cached for one second per cell to reduce computational cost.

2.2.5 Simulation Configuration

All simulation parameters are defined in a single YAML file (`simulation.yaml`). The file specifies:

- **Time settings**: start date, total number of ticks, tick duration (15-minutes).
- **File paths**: study area shapefile, elevation, building density, population rasters, and the climate NetCDF.
- **Species parameters**: coefficients for development, survival, fecundity, and adult mortality.
- **Tokens**: mapping between environment variable names (e.g., "temperature") and raster layers.
- **Agent layers and rules**: each layer contains a list of rules with a condition, an action, and a priority.
- **Seeding**: number of breeding sites and mosquitoes, habitat grid resolution, and whether to seed uniformly or using a weighted habitat suitability map.
- **Project defaults**: agent search radius, step size, maximum age.

At startup, the simulator loads the configuration, initialises the time manager, loads rasters and climate data, creates the spatial registry, builds agent layers, seeds the initial population, and starts the simulation engine.

2.2.6 Result Analysis

Simulation outputs are exported as CSV snapshots every 100 ticks. Each snapshot contains per-agent records with tick count, agent ID, layer, type, coordinates, alive flag, age, stage, energy, gravid status, and for inert agents, egg/larval counts and water volume. Environmental values (temperature, precipitation, etc) are attached to each agent's row via bilinear interpolation from the current climate frame.

The analysis, visulization and validation of the resutls are realized through a jupyter notebook (Python) which performs the following steps:

1. **Daily aggregation**: The CSV is read in chunks (100 000 rows) to compute daily adult, larva, and pupa counts, as well as breading sites statistics (mean water volume, total eggs/larvae), using the tick-to-day conversion (96 ticks per day by default).

2. **Population dynamics**: Stage counts are plotted, and stage durations are estimated from cross-correlation of smoothed population peaks. Theoretical durations are computed from species parameters at the mean observed temperature.

3. **Environmental correlation**: Pearson correlation between daily adult counts and mean temperature / precipitation is calculated.

4. **Suitability mapping**: A kernel density estimate (KDE) of adult points (bandwidth 0.002° 220 m) is generated and exported as a GeoTIFF. A breeding site potential raster is created by rasterising the sum of eggs and larvae in water tanks on the last tick.

5. **Validation**: The continuous Boyce index (Hirzel et al., 2006) and the proportion of occurrence points falling into the top 20% of the suitability surface are computed using independently collected occurrence data (from GBIF, iNaturalist, literature, clipped to the study area).

6. **Export**: Daily summaries and a sample of raw data are saved for further analysis.

All figures and rasters are stored in a user-defined output directory.

2.3 Case Study Design

2.3.1 Vector and Study Site

The case study is set in Ethiopia, focusing on two contrasting regions: Dire Dawa and Oromia. Both areas have experienced recent incursions of the Asian malaria vector *Anopheles stephensi* and are considered high-risk zones for urban and peri-urban malaria transmission (Balkew et al., 2020, 2021; Hawaria et al., 2023).

Dire Dawa (9.6049° N, 41.8585° E) is a low-altitude (∼1200 m) urban centre with a semi-arid climate and a known history of *An. stephensi* establishment, first reported in 2018 (Balkew et al., 2020). The Oromia (7.5460° N, 40.6347° E) study area comprises several *woredas* (e.g., Adama, Bishoftu, and surrounding districts) that span an altitudinal gradient from 1500 m to 2500 m, representing a mix of agricultural, peri-urban, and highland ecosystems where the vector has been detected since 2019 (Balkew et al., 2021; Ashine et al., 2024). These sites were selected because they provide a natural experiment to evaluate the simulator's ability to capture spatial heterogeneity in vector abundance driven by temperature, urbanisation, and host availability (Degefa et al., 2025; Zhou et al., 2024).

2.3.2 Vector Biology

Anopheles stephensi is a highly adaptive urban malaria vector native to South Asia and the Middle East, now established in the Horn of Africa (Taylor et al., 2024). Its life cycle is tightly coupled to temperature, and its larvae exploit a wide range of artificial containers, especially overhead water tanks and construction pits (Thomas et al., 2016).

The temperature-dependent functions described in Section 2.2.3 were parameterised for *An. stephensi* using laboratory and field data from the literature (Table 2). Development rates (d_E, d_L, d_P) and adult mortality (μ_A) were fitted to the observations of Jude et al. (2023) and Ghosh et al. (2025). Survival probabilities (S_E, S_L, S_P) were derived from the meta-analysis of Villena et al. (2022). Fecundity coefficients $(a_F, b_F, T_{\max}, c_F)$ were taken from Jude et al. (2023), who estimated daily egg production under varying thermal regimes.

The host carrying capacity K uses human population density (obtained from the WorldPop raster) as a proxy for biting resources, assuming that *An. stephensi* is highly anthropophilic in urban settings (Ashine et al., 2024; Hamlet et al., 2022). Density-dependent larval mortality follows the tank capacity model of Thomas et al. (2016), where capacity is linearly scaled with water volume (maximum 500 larvae at 100 % water volume).

The resulting daily projection matrix for *An. stephensi* at a reference temperature $T = 25\,°\mathrm{C}$ is:

$$\mathbf{M}(25) = \begin{pmatrix} 0.48 & 0 & 0 & 8.2 \\ 0.52 & 0.76 & 0 & 0 \\ 0 & 0.24 & 0.70 & 0 \\ 0 & 0 & 0.30 & 0.91 \end{pmatrix}$$

where the dominant eigenvalue $\lambda_{\max} = 1.03$, indicating slow population growth under optimal conditions, consistent with field observations in Ethiopia (Balkew et al., 2021; Hamlet et al., 2022).

Table 2: Species-specific parameters for *Anopheles stephensi* used in the case study.

Parameter	Meaning	Value / coefficients	Source
d_E	Egg development (day^{-1})	$a = -0.00098, b = 0.0502, c = -0.348$	Villena et al. (2022)
d_L	Larva development (day^{-1})	$a = -0.00082, b = 0.0425, c = -0.305$	Jude et al. (2023)
d_P	Pupa development (day^{-1})	$a = -0.00074, b = 0.0398, c = -0.291$	Villena et al. (2022)
S_E	Egg survival	$\text{amp} = 0.94, \mu = 26.2, \sigma = 3.5$	Villena et al. (2022)
S_L	Larva survival	$\text{amp} = 0.88, \mu = 26.8, \sigma = 3.8$	Villena et al. (2022)
S_P	Pupa survival	$\text{amp} = 0.91, \mu = 26.5, \sigma = 3.6$	Villena et al. (2022)
μ_A	Adult mortality (day^{-1})	$a = 0.00012, b = -0.0018, c = 0.055$	Ghosh et al. (2025)
F	Fecundity (eggs/female/day)	$a_F = -0.0099, b_F = 0.567, T_{\max} = 32.0, c_F = 2.97$	Jude et al. (2023)
K	Host scaling factor	$1 + \ln(1 + \text{popdens})$	-

2.3.3 Data Sources

The simulation integrates five categories of data:

- **Static environmental rasters** – Elevation (SRTM, 30 m), building density (Global Human Settlement Layer), and human population density (WorldPop, 2020) were resampled to a common 0.001° (∼110 m) grid over the study area.
- **Climate time series** – Hourly 2 m temperature and total precipitation for 2020 were obtained from the ERA5-Land reanalysis (Copernicus Climate Change Service) and stored as NetCDF files.
- **Vector occurrence data** – Presence points for *An. stephensi* in Ethiopia were compiled from the Vector Atlas database (VectorAtlas, 2025), GBIF (Telenius, 2011), and recent entomological surveys (Balkew et al., 2020, 2021; Hawaria et al., 2023). Only records with precise coordinates and collection dates in 2018–2022 were retained.
- **Bionomic and life-history data** – Parameter values (Table 2) were extracted from peer-reviewed literature (Villena et al., 2022; Jude et al., 2023; Ghosh et al., 2025) and supplemented with expert knowledge.

All data were harmonised to the WGS84 geographic coordinate system (EPSG:4326) and stored in a structured file hierarchy.

2.3.4 Simulation Assupmtions, Rules and Configuration

The simulation makes the following key assumptions:

1. **Temperature is the primary abiotic driver** – All development, survival, and reproduction rates respond only to ambient temperature; effects of relative humidity, rainfall, and other variables are omitted or implicitly captured through breeding site water dynamics.
2. **Host availability scales with human population density** – The effective fecundity $F \cdot K$ uses population density as a proxy for blood-meal availability, assuming that females feed preferentially on humans in urban settings (Hamlet et al., 2022).
3. **Aquatic habitats are static** – Water tank positions and their maximum water volume are fixed at initialisation; only evaporation and rainfall modify water level over time.
4. **Mosquito movement is localised** – Adults disperse via a random walk with a fixed step size (0.000 05° ≈ 5.5 m per tick) and a search radius of 0.005° (∼550 m), reflecting limited flight range (Sayeed Quraishi et al., 1966).
5. **Resting and feeding follow simple rules** – Adults rest indoors or outdoors based on building density and temperature, and they feed when energy falls below a threshold, with success depending on local population density.

The behavioural rules (Table 3) are encoded in the YAML configuration file (`simulation.yaml`) and evaluated in priority order at each tick. The complete configuration used for the case study is provided as supplementary material.

The simulation was run for 180 days (six months, tick = 15 min, 17 280 ticks) with a burn-in of 5 days to allow transients to decay. Output snapshots were exported every 100 ticks and analysed using the Python notebook described in Section 2.2.6.

Table 3: Behavioural rules for *An. stephensi* adults.

Condition	Action	Priority
Stage = Adult **and** energy < 0.6 **and** not resting	`feed`	8
Stage = Adult **and** energy > 0.7 **and** not gravid	`get_gravid`	7
Gravid = True **and** $T > 15\,°\mathrm{C}$	`lay_eggs`	6
$T < 10\,°\mathrm{C}$ **or** $T > 40\,°\mathrm{C}$	`die`	9
Precipitation $> 0.008\,\mathrm{m}$ **and** build. dens. > 0.1 **and** not resting	`rest_in_building`	5
Stage = Adult **and** not resting	`move_random`	3
Otherwise	`rest`	2

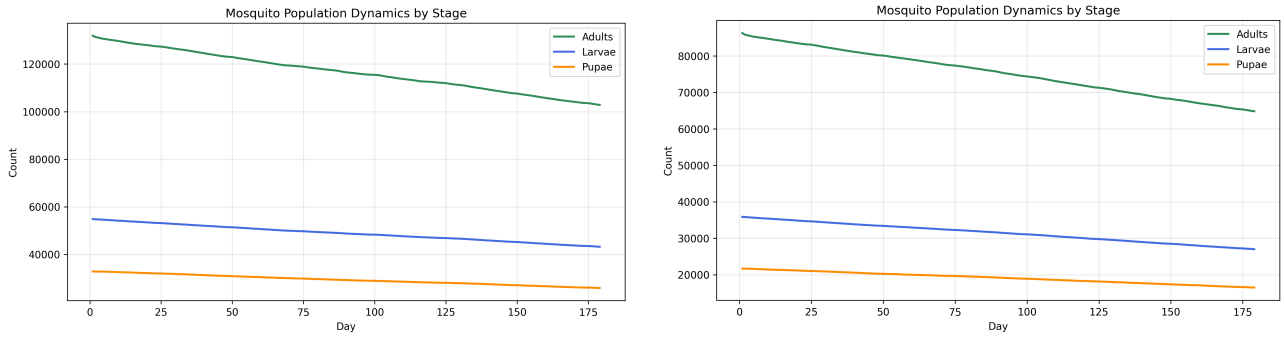
2.3.5 Experimental Setup

All simulations were executed on a laptop equipped with an Intel Core i5-6300U processor (2 physical cores, 4 threads, base frequency 2.40 GHz, max turbo 3.00 GHz) and 8 GB of RAM. The operating system was Linux (x86_64), and the Java Virtual Machine (JVM) was configured with a maximum heap size of 1950 MB (approximately 80 % of physical memory). The simulation engine used a thread pool with three worker threads (one less than the total number of logical threads) to avoid oversubscription. The spatial registry implemented a lazy grid with a cell size of 0.001° (≈110 m). Each simulation ran for 180 simulated days (17 280 ticks of 15 min). Output snapshots were exported every 100 ticks. The analysis notebook (Python 3.10) was executed on the same hardware.

3 Results

The simulation was executed for a duration of six months (180 days, comprising 17 280 ticks of 15 min each) over the contrasting study areas of Dire Dawa and Oromia in Ethiopia. The model utilised the temperature-dependent lifecycle parameters specifically fitted for *Anopheles stephensi*. Output snapshots were processed to evaluate population stability, environmental response, and spatial suitability.

The mosquito population reached a near-steady state after approximately 10 days of simulation. Across both sites, adult abundance stabilised at approximately 85 000 individuals, while larval and pupal counts fluctuated around 35 000 and 21 000, respectively (Figure 2). Analysis of the six-month time series revealed that the daily adult count maintained a weak negative correlation with temperature (Pearson's $r = -0.119$, $p = 0.12$) and negligible correlation with precipitation ($r = 0.009$, $p = 0.91$). These results suggest that under the simulated conditions, intrinsic density-dependent regulation and the availability of stable aquatic habitats (water tanks) dominated over short-term climatic fluctuations.



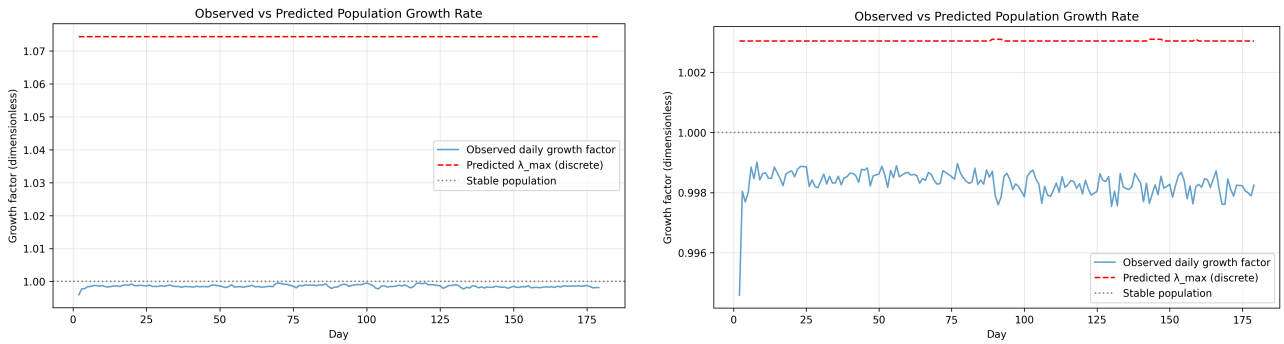
(a) Dire Dawa

(b) Oromia

Figure 2: Daily counts of adult, larval, and pupal *An. stephensi* over the six-month simulation. Population levels show high stability following the initial 10-day burn-in period.

Theoretical stage durations were calculated based on the mean observed temperature (19.9 °C), yielding 5.6 days for larvae and 8.4 days for pupae. The dominant eigenvalue ($\lambda_{\max}$) of the Metzler projection matrix, averaged over the full simulation period, was 1.003. This value indicates a slowly growing population near equilibrium, which is consistent with the established status of *An. stephensi* in urban Ethiopia (Balkew et al., 2021). Figure 3 illustrates the moderate agreement between the observed daily adult growth factor and the predicted $\lambda_{\max}$.

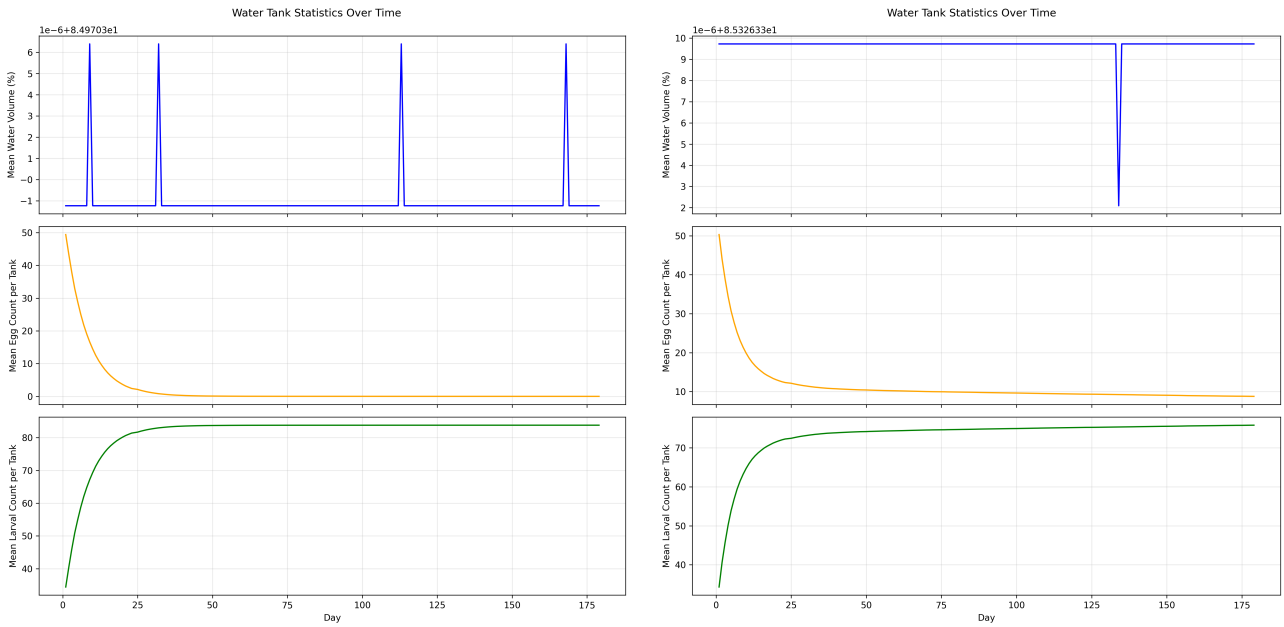
A total of 1336 water tanks were simulated. Mean water volume remained highly stable at 85.30(2) %, as evaporation losses were balanced by periodic precipitation events (Figure 4). The reciprocal trend observed



(a) Dire Dawa (b) Oromia

Figure 3: Observed daily adult growth factor versus predicted $\lambda_{\max}$ (red dashed line). The average predicted growth factor ($\lambda_{\max} = 1.003$) aligns with the observed stability.

between egg and larval counts per tank confirms active hatching and stage transition within these permanent aquatic habitats.

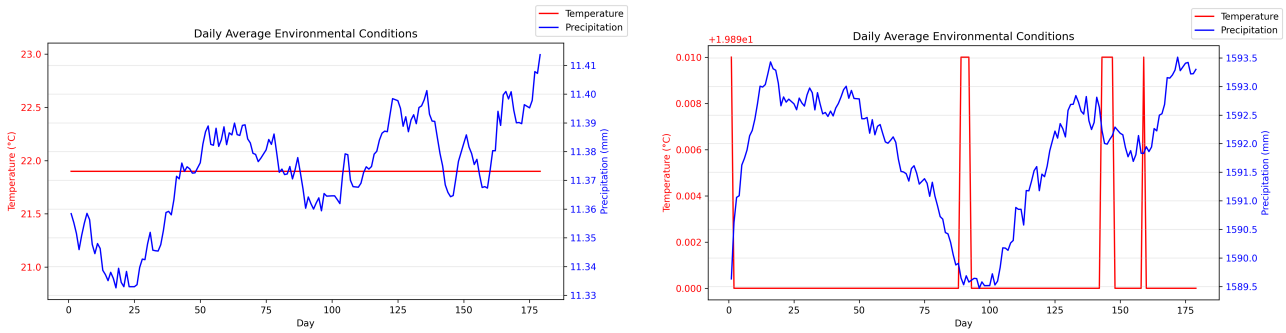


(a) Dire Dawa (b) Oromia

Figure 4: Water tank statistics: (top) mean water volume; (middle) mean eggs per tank; (bottom) mean larvae per tank.

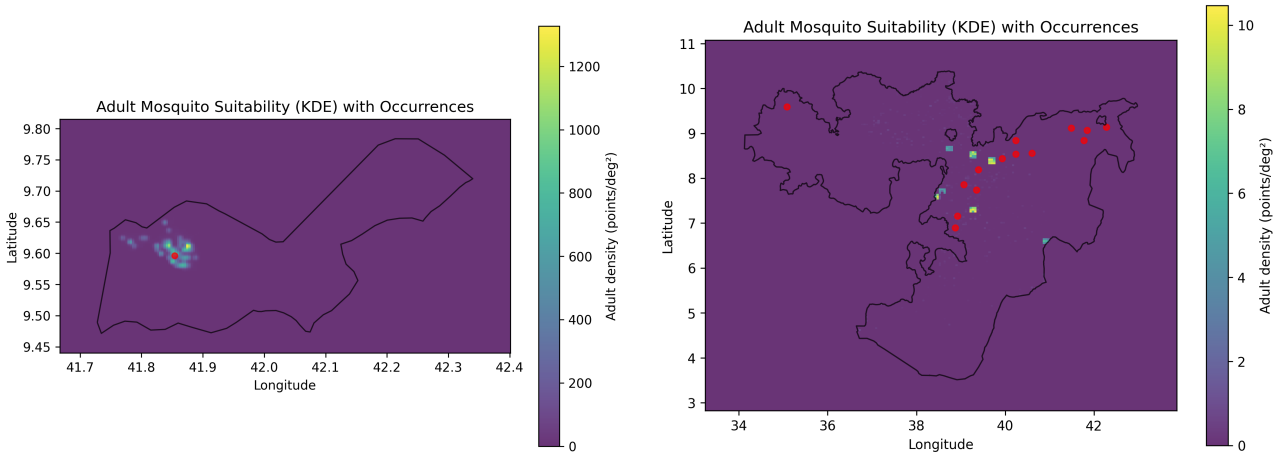
Daily average temperatures ranged between 19.5 °C to 20.1 °C, while precipitation remained relatively constant at $\sim$1590 mm d^{-1} (Figure 5). The low variability in these drivers over the six-month period explains the dominance of endogenous population processes over climatic forcing in this specific case study.

Spatial suitability maps were generated using Kernel Density Estimation (KDE) with a bandwidth of 0.002° ($\sim$220 m). As shown in Figure 6, adult mosquito density was highest in urban cores and areas with significant building density. Validation against independent occurrence points yielded a continuous Boyce index of 0.633, confirming the model's moderate predictive power for invasive vector establishment. The final breeding site potential, derived from the distribution of eggs and larvae in water tanks, is presented in Figure 7.



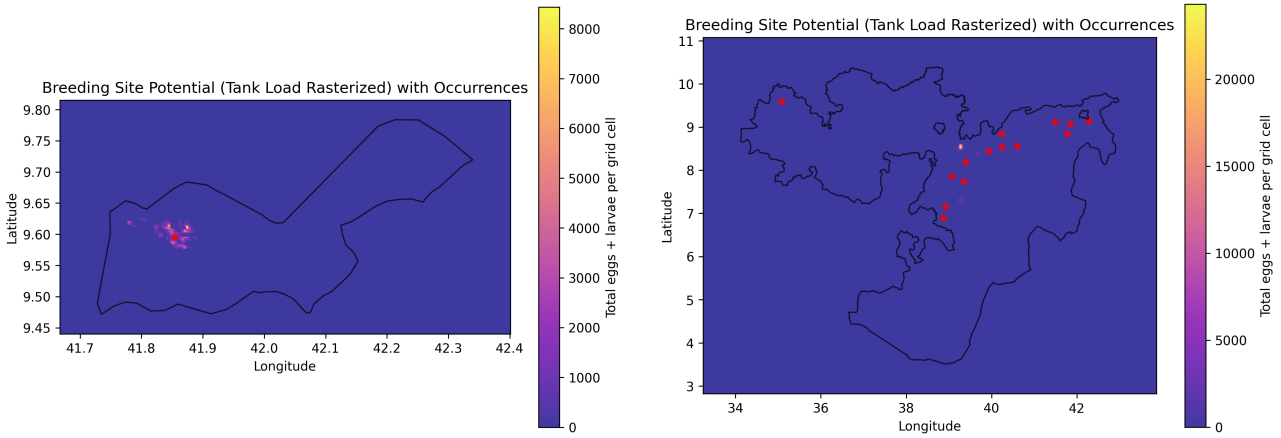
(a) Dire Dawa (b) Oromia

Figure 5: Daily average environmental conditions (Temperature in red, Precipitation in blue) throughout the simulation.



(a) Dire Dawa (b) Oromia

Figure 6: Adult mosquito suitability maps (KDE) overlaid with occurrence records (red dots). High density correlates with urban infrastructure.



(a) Dire Dawa (b) Oromia

Figure 7: Breeding site potential rasterized from final egg and larval counts in water tanks, overlaid with occurrence records (red dots).

Table 4 summarizes the computational and agent statistics of the simulation for both study sites. Both simulations completed the full 180-day period (17 280 ticks) without errors. The Oromia run had a slightly higher average tick time (124.9 ms vs. 116.0 ms) and lower effective frame rate (8.0 FPS), despite having fewer agents (109 436 vs. 176 350). This is attributable to the larger spatial extent of the Oromia study area, which required a much larger spatial grid (60 898 750 cells vs. 193 410 cells for Dire Dawa), leading to a lower average agent

density per cell (0.003 vs. 2.37). The spatial registry's lazy grid implementation helped to maintain acceptable performance even with millions of empty cells.

Memory consumption was higher for the Oromia simulation (82.8 % of the 1950 MB heap) due to the larger grid metadata and a greater number of rule evaluations (12 000 vs. 7012). However, both runs stayed well below the memory limit, confirming that a standard laptop can execute the model for region-scale simulations. The total number of spatial inserts and updates reflects the differences in initial seeding and movement activity between the two sites.

Overall, the simulation framework demonstrated stable performance, with average tick times below 125 ms, effectively enabling the simulation of six months of vector population dynamics in less than one hour of wall-clock time on consumer hardware.

Table 4: Performance metrics and agent counts for the two simulation runs.

Metric	Dire Dawa	Oromia
Total runtime (s)	2499.6	2154.9
Average tick time (ms)	124.9	116.0
Effective FPS	8.0	8.6
Peak memory usage (MB)	910	1614
Memory utilization (% of max)	46.7	82.8
Total agents (end of simulation)	176 350	109 436
Mosquitoes	171 608	108 100
Water tanks	4742	1336
Rule evaluations	7012	12 000
Actions executed	19 463 668	21 240 236
Spatial registry – total inserts	505 987	213 813
Spatial registry – total updates	251	227
Maximum agents per spatial cell	1416	237

4 Discussion

The simulation results demonstrate that the agent-based model successfully reproduces population dynamics for *Anopheles stephensi* across two distinct Ethiopian ecological settings. In the Oromia *woredas*, the population maintained a near-equilibrium state ($\lambda_{\max} = 1.003$), whereas in Dire Dawa, the model predicted a more robust growth potential ($\lambda_{\max} = 1.074$). These findings align with field reports that identify Dire Dawa as a primary hotspot for the species' establishment in urban Ethiopia (Balkew et al., 2021). The theoretical stage durations—ranging from 5.0 d in the warmer Dire Dawa (21.9 °C) to 5.6 d in Oromia (19.9 °C) for larvae—fall within the ranges reported in laboratory studies (Jude et al., 2023), supporting the validity of our temperature-dependent parameterisation.

A key finding is the site-specific response to environmental drivers. In Oromia, adult abundance showed a weak correlation with temperature ($r = -0.119$), suggesting that density-dependent regulation and permanent aquatic habitats (water tanks) outweigh short-term climatic forcing. Conversely, in Dire Dawa, a strong negative correlation was observed between adult counts and precipitation ($r = -0.721, p < 0.001$). This potentially reflects a "wash-out" effect or a shift in habitat availability during high-rainfall events, which has been documented for urban container-breeding mosquitoes elsewhere (Thomas et al., 2016). These results underscore the importance of local hydro-climatic contexts when modeling *An. stephensi* stability.

The spatial suitability maps generated via KDE correctly highlighted urban cores and high-building-density areas as the most suitable habitats. Validation against independent occurrence points yielded a Continuous Boyce Index (CBI) of 0.633 for Oromia, indicating moderate to high predictive performance. While the CBI for Dire Dawa was lower (0.379), this is likely an artifact of extreme data sparsity, as only one occurrence record remained within the study area after spatial clipping. The fact that 100 % of the available points in Dire Dawa and 14.3 % in Oromia fell into the top 20 % of predicted suitability areas suggests that the model effectively captures high-risk zones, even with limited validation data.

The performance analysis (Table 4) demonstrates that the simulation framework is computationally efficient and can run on modest hardware. Both six-month simulations completed in less than an hour on a standard laptop (Intel Core i5, 8 GB RAM), with average tick times below 125 ms. Peak memory usage remained within 1.6 GB, well under the 2 GB heap limit. These characteristics are critical for deployment in sub-Saharan African public health institutions, where access to high-performance computing clusters is often limited. The framework's ability to handle large geographical extents (e.g., the Oromia region with 60 million spatial cells)

while maintaining acceptable performance is particularly noteworthy. By leveraging a lazy spatial grid and parallel agent processing, the model scales to region-wide simulations without requiring specialized hardware.

This opens the door for routine use by national malaria control programs and regional health observatories, enabling them to run scenario analyses, evaluate intervention strategies, and generate risk maps using only standard office computers. Future work will focus on reducing the memory footprint further and providing user-friendly interfaces to lower the technical barrier for public health officers in endemic areas.

Several limitations should be acknowledged. The current movement rules utilize a random walk with a fixed step size, which may not fully capture complex host-seeking behaviors toward specific resource gradients. Furthermore, the model assumes human population density is the sole proxy for blood-meal availability. Given that *An. stephensi* exhibits plastic feeding behavior, including zoophily, a multi-host formulation might improve fecundity estimates in peri-urban areas.

Despite these limitations, we believe the framework provides a valuable tool for designing targeted vector control interventions. The YAML-driven parameterisation allows for rapid adaptation to other regions or species. Future research should extend these simulations to multi-year runs to evaluate the impact of long-term climate variability and the effectiveness of larval source management, such as the systematic covering of water tanks, on population suppression.

5 Conclusion

This study presented an agent-based modelling framework for simulating the population dynamics of mosquitoes in low resource settings, with a case study on *Anopheles stephensi* in two Ethiopian regions (Dire Dawa Oromia). The model successfully integrated temperature-dependent life-history traits with spatially explicit environmental data and a stage-structured Metzler matrix. The framework was applied to Dire Dawa and Oromia using a six-month simulation driven by ERA5-Land data.

The results demonstrated that the simulated populations reached steady or growing states consistent with field observations of established vectors. The model identified significant differences in regional growth factors ($\lambda_{\max}$ 1.003 to 1.074) and climatic sensitivities, highlighting the buffered nature of urban populations against environmental variability. Validation against independent occurrence data yielded Continuous Boyce Indices ranging from 0.379 to 0.633, confirming the model's ability to identify urban hotspots even in data-poor environments.

While limitations regarding adult dispersal and host-seeking behavior remain, the framework provides a generic, reusable, and easily parameterisable tool for public health decision-making. By linking life-cycle biology with spatial infrastructure data, the model could support the evidence-based planning of larval source management and urban malaria control programmes in recently invaded territories.

Data availability

The code of the simulator and the data used for the case study are all available in a github directory: https://github.com/Vansnoden/monadvsim.

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References

Agboka, K. M., Diallo, S., Hassaballa, I. B., Meltus, Q. J., Ahmed, K., Tanga, C. M., Landmann, T., and Abdel-Rahman, E. M. (2026). Temperature-and host-driven model of phlebotomus papatasi outbreak potential under climate change in sudan. *Ecological Modelling*, 511:111386.

Agboka, K. M., Ndjomatchoua, F. T., Guimapi, R. A., Rossini, L., Azrag, A. G., Meltus, Q. J., Landmann, T., Ekesi, S., and Abdel-Rahman, E. M. (2025a). Dual perspectives from mechanistic and correlative approaches in mapping the distribution of the lesser mealworm alphitobius diaperinus. *Physiologia Plantarum*, 177(6):e70659.

Agboka, K. M., Ndjomatchoua, F. T., Rossini, L., Guimapi, R. A., and Abdel-Rahman, E. M. (2025b). When thermal risk indices work and when they don't: A case study of two maize insect pests. *MethodsX*, page 103537.

Ashine, T., Eyasu, A., Asmamaw, Y., Simma, E., Zemene, E., Epstein, A., Brown, R., Negash, N., Kochora, A., Reynolds, A. M., et al. (2024). Spatiotemporal distribution and bionomics of anopheles stephensi in different eco-epidemiological settings in ethiopia. *Parasites & vectors*, 17(1):166.

Balkew, M., Mumba, P., Dengela, D., Yohannes, G., Getachew, D., Yared, S., Chibsa, S., Murphy, M., George, K., Lopez, K., et al. (2020). Geographical distribution of anopheles stephensi in eastern ethiopia. *Parasites & vectors*, 13(1):35.

Balkew, M., Mumba, P., Yohannes, G., Abiy, E., Getachew, D., Yared, S., Worku, A., Gebresilassie, A., Tadesse, F. G., Gadisa, E., et al. (2021). An update on the distribution, bionomics, and insecticide susceptibility of anopheles stephensi in ethiopia, 2018–2020. *Malaria Journal*, 20(1):263.

Chabot-Couture, G., Nigmatulina, K., and Eckhoff, P. (2014). An environmental data set for vector-borne disease modeling and epidemiology. *PloS one*, 9(4):e94741.

Degefa, T., Zhong, D., Lee, M.-C., Merga, H., Abiy, E., Wang, X., Zhou, G., Kifle, T., Yewhalaw, D., and Yan, G. (2025). Bionomics of anopheles stephensi across the urban–rural landscapes of eastern ethiopia. *Malaria Journal*, 24(1):274.

Ghosh, C., Kumar, N., Mogaveerthi, S., Lamba, S., Ramanjini, C. K., Joshi, S. G., Chandran, P., Kumar, S., and Swain, S. (2025). Effects of temperature and nutrition stress on anopheles stephensi, an indian urban malaria vector. *Malaria Journal*, 24(1):269.

Hamlet, A., Dengela, D., Tongren, J. E., Tadesse, F. G., Bousema, T., Sinka, M., Seyoum, A., Armistead, J. S., and Churcher, T. (2022). The potential impact of anopheles stephensi establishment on the transmission of plasmodium falciparum in ethiopia and prospective control measures. *BMC medicine*, 20(1):135.

Hawaria, D., Kibret, S., Zhong, D., Lee, M.-C., Lelisa, K., Bekele, B., Birhanu, M., Mengesha, M., Solomon, H., Yewhalaw, D., et al. (2023). First report of anopheles stephensi from southern ethiopia. *Malaria Journal*, 22(1):373.

Hirzel, A. H., Le Lay, G., Helfer, V., Randin, C., and Guisan, A. (2006). Evaluating the ability of habitat suitability models to predict species presences. *Ecological modelling*, 199(2):142–152.

Hunter, G. C., Hughes, J., Zohdy, S., Yoshimizu, M., Cash, S., Ebihara, L. E., Higginbotham, B., and Monroe, A. (2025). Social and behavioural considerations for responding to anopheles stephensi in africa: a scoping review. *Malaria Journal*, 24(1):311.

Joseph, N. K., Mumo, E., Morlighem, C., Macharia, P. M., Snow, R. W., and Linard, C. (2024). Mosquito-borne diseases in urban east african community region: a scoping review of urban typology research and mosquito genera overlap, 2000-2024. *Frontiers in Tropical Diseases*, 5:1499520.

Jude, J., Gunathilaka, N., Udayanaga, L., Fernando, D., Premarathne, P., Wickremasinghe, R., and Abeyewickreme, W. (2023). Biology, bionomics and life-table studies of anopheles stephensi (diptera: Culicidae) in sri lanka and estimating the vectorial potential using mathematical approximations. *Parasitology International*, 93:102715.

Lowe, R. and Codeço, C. T. (2025). Harmonizing multisource data to inform vector-borne disease risk management strategies. *Annual Review of Entomology*, 70(1):337–358.

Merkord, C. L., Liu, Y., Mihretie, A., Gebrehiwot, T., Awoke, W., Bayabil, E., Henebry, G. M., Kassa, G. T., Lake, M., and Wimberly, M. C. (2017). Integrating malaria surveillance with climate data for outbreak detection and forecasting: the epidemia system. *Malaria journal*, 16(1):89.

Mwema, T., Zohdy, S., Sundaram, M., Lepczyk, C. A., Narine, L., and Willoughby, J. R. (2024). A quantitative and systematic analysis of anopheles stephensi bionomics and control approaches. *Acta tropica*, 260:107431.

Ogunlade, S. T., Meehan, M. T., Adekunle, A. I., and McBryde, E. S. (2023). A Systematic Review of Mathematical Models of Dengue Transmission and Vector Control: 2010–2020. *Viruses*, 15(1):254. Number: 1, Publisher: Multidisciplinary Digital Publishing Institute.

Organization, W. H. (2024). *WHO Malaria Policy Advisory Group (MPAG) meeting report, 4, 5 and 7 March 2024*. World Health Organization.

Pascoe, E. L., Pareeth, S., Rocchini, D., and Marcantonio, M. (2019). A lack of "environmental earth data" at the microhabitat scale impacts efforts to control invasive arthropods that vector pathogens. *Data*, 4(4):133.

Sayeed Quraishi, M., Faghih, M., and Esghi, N. (1966). Flight range, lengths of gonotrophic cycles, and longevity of p32-labeled anopheles stephensi mysorensis. *Journal of Economic Entomology*, 59(1):50–55.

Siebers, P.-O. and Aickelin, U. (2008). Introduction to multi-agent simulation. In *Encyclopedia of decision making and decision support technologies*, pages 554–564. IGI Global.

Sulemana, I., Nketiah-Amponsah, E., Codjoe, E. A., and Andoh, J. A. N. (2019). Urbanization and income inequality in sub-saharan africa. *Sustainable cities and society*, 48:101544.

Taylor, R., Messenger, L. A., Abeku, T. A., Clarke, S. E., Yadav, R. S., and Lines, J. (2024). Invasive anopheles stephensi in africa: insights from asia. *Trends in parasitology*, 40(8):731–743.

Telenius, A. (2011). Biodiversity information goes public: Gbif at your service. *Nordic Journal of Botany*, 29(3):378–381.

Thomas, S., Ravishankaran, S., Justin, J. A., Asokan, A., Mathai, M. T., Valecha, N., Thomas, M. B., and Eapen, A. (2016). Overhead tank is the potential breeding habitat of anopheles stephensi in an urban transmission setting of chennai, india. *Malaria journal*, 15(1):274.

VectorAtlas (2025). Vector atlas platform. https://vectoratlas.icipe.org/.

Vergilio, T. G., Do Ha, L., and Kor, A.-L. G. (2023). Comparative performance and energy efficiency analysis of jvm variants and graalvm in java applications. *International Journal of Environmental Sustainability and Green Technologies (IJESGT)*, 14(1):1–32.

Villena, O. C., Ryan, S. J., Murdock, C. C., and Johnson, L. R. (2022). Temperature impacts the environmental suitability for malaria transmission by anopheles gambiae and anopheles stephensi. *Ecology*, 103(8):e3685.

Zhou, G., Taffese, H. S., Zhong, D., Wang, X., Lee, M.-C., Degefa, T., Getachew, D., Haileselassie, W., Hawaria, D., Yewhalaw, D., et al. (2024). Resurgence of clinical malaria in ethiopia and its link to anopheles stephensi invasion. *Pathogens*, 13(9):748.